



ME 523: Product Design

Course Code	ME 523
Course Name	Product Design
L-T-P-C	3-0-0-3, Elective
Prerequisites	None
Intended for	B.Tech./M.Tech/Ph.D.
Mutual Exclusion	None
Approval	50th BoA (Jul 2023)

Course Contents

- **Introduction, generic development process, opportunity identification:** Characteristics of Successful Product Development, Generic Product Development Process, Concept development, Generic product development process, Opportunity structure and the associate process. [10 Hours]
- **Product planning, customer needs and product specification:** Product planning process (Identification, Evaluation, Allocation), Importance of latent needs, Customer needs identification, Target specifications, Final product specifications [09 Hours]
- **Concept generation, selection and testing:** Activity of concept generation, Five step method, Choosing a concept, Concept screening and scoring, Concept tests, Survey population and format, response measurement and reflecting on the results [09 Hours]
- **Product architecture, Industrial design, Design for environment and economics:** Product Architecture, Modularity, Implications of the Architecture, Establishing the Architecture, Assessing the Need for Industrial Design, Impact of Industrial Design, Quality assessment, Design for environment and associated process, Elements of Economic Analysis, Economic analysis process, Patents and IPR, Case studies across all the disciplines [14 Hours]

Text Books

1. K T Ulrich and S D Eppinger, **Product Design and Development**, McGraw Hill, 2000.
2. K Otto and K Wood, **Product Design**, Pearson Education, Inc. 2001
3. K G Cooper, **Rapid Prototyping Technology**, Marcel Dekker, Inc. 2001
4. D T Pham and S S Dimov, **Rapid Manufacturing**, Springer-Verlag, 2001